MTH100 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

https://chat.whatsapp.com/Hrh5CrjNO6u31kFiNX352H

MTH100 EXAMPLES

Topic 6 (Coordinate Geometry)

Difficult Example 1 (Distance Between Two Points)

Question:
Find the distance between the points **A(-8, 5)** and **B(4, -7).**

Solution

Using the distance formula,

$$
AB=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}
$$

Substitute the values:

$$
=\sqrt{(4-(-8))^2+(-7-5)^2}
$$

$$
=\sqrt{12^2+(-12)^2}
$$

$$
=\sqrt{144+144}
$$

$$
=\sqrt{288}
$$

$$
=12\sqrt2
$$

Answer: $12\sqrt2$

VULMS Mastery

Difficult Example 2 (Midpoint)

Question:
Find the midpoint of the line joining **(-9, 14)** and **(7, -10).**

Solution

Midpoint formula:

https://chat.whatsapp.com/Hrh5CrjNO6u31kFiNX352H

$$
M\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)
$$

$$
M=\left(\frac{-9+7}{2},\frac{14+(-10)}{2}\right)
$$

$$
M=\left(\frac{-2}{2},\frac4{2}\right)
$$

$$
M=(-1,2)
$$

Answer: (-1, 2)

Difficult Example 3 (Slope)

Question:
Find the slope of the line passing through **(-6, 9)** and **(8, -5)**.

Solution

$$
m=\frac{y_2-y_1}{x_2-x_1}
$$

$$
=\frac{-5-9}{8-(-6)}
$$

$$
=\frac{-14}{14}
$$

$$
=-1
$$

Answer: −1

Difficult Example 4 (Equation of Line)

Question:
Find the equation of the line passing through **(5,-2)** having slope **3**.

Solution

VULMS Mastery

https://chat.whatsapp.com/Hrh5CrjNO6u31kFiNX352H

Point-slope form:

[
$y-y_1=m(x-x_1)$
]

[
$y+2=3(x-5)$
]

[
$y+2=3x-15$
]

[
$y=3x-17$
]

Answer:

[
$\boxed{y=3x-17}$
]

Difficult Example 5 (Slope from Equation)

Question:
Find the slope of the line

[
$8x+5y=20$
]

Solution

Write in slope-intercept form.

[
$5y=-8x+20$
]

[
$y=-\frac85x+4$
]

Hence,

[
$m=-\frac85$
]

VULMS Mastery

Answer: (-\frac85)

Difficult Example 6 (Intersection of Two Lines)

Question:
Find the point of intersection of

[
$2x+3y=12$
]

and

[
$4x-y=10$
]

Solution

From second equation,

[
$y=4x-10$
]

Substitute into first equation.

[
$2x+3(4x-10)=12$
]

[
$2x+12x-30=12$
]

[
$14x=42$
]

[
$x=3$
]

Now,

[
$y=4(3)-10=2$
]

Answer: (3,2)

VULMS Mastery

https://chat.whatsapp.com/Hrh5CrjNO6u31kFiNX352H

Difficult Example 7 (Parallel or Perpendicular)

Question:
Determine whether the following lines are parallel, perpendicular or neither.

$$6x-4y=8$$

$$2x+3y=9$$

Solution

First line:

$$-4y=-6x+8$$

$$y=\frac32x-2$$

Slope,

$$m_1=\frac32$$

Second line:

$$3y=-2x+9$$

$$y=-\frac23x+3$$

Slope,

$$m_2=-\frac23$$

Now,

VULMS Mastery

https://chat.whatsapp.com/Hrh5CrjNO6u31kFiNX352H

$$
m_1m_2=\frac32\times-\frac23=-1
$$

Therefore,

The lines are **perpendicular**.

Answer: Perpendicular

Difficult Example 8 (Equation of Circle)

Question:
Find the center and radius of

$$
(x-7)^2+(y+5)^2=81
$$

Solution

Compare with

$$
(x-h)^2+(y-k)^2=r^2
$$

Here,

$$
h=7,\quad k=-5,\quad r=\sqrt{81}=9
$$

Answer:

Center = **(7,-5)**

Radius = **9**

Difficult Example 9 (Completing the Square)

Question:
Find the center and radius of

$$
x^2+y^2-8x+10y-20=0
$$

VULMS Mastery

https://chat.whatsapp.com/Hrh5CrjNO6u31kFiNX352H

Solution

Move constant.

$$x^2-8x+y^2+10y=20$$

Complete the square.

$$(x^2-8x+16)+(y^2+10y+25)=20+16+25$$

$$(x-4)^2+(y+5)^2=61$$

Therefore,

Center = **(4,-5)**

Radius

$$r=\sqrt{61}$$

Answer:

Center = **(4,-5)**

Radius = **($\sqrt{61}$)**

VULMS Mastery

Difficult Example 10 (Combined Concept)

Question:
The endpoints of a diameter of a circle are

$$A(-6,-2),\qquad B(8,10)$$

Find

1. The center of the circle.

2.

3.

The radius.

4.
5.

The equation of the circle.

6.

Solution

Step 1: Center

The center is the midpoint.

$$\left(\frac{-6+8}{2},\frac{-2+10}{2}\right)$$

$$=(1,4)$$

Step 2: Diameter

$$AB=\sqrt{(8+6)^2+(10+2)^2}$$

$$=\sqrt{14^2+12^2}$$

$$=\sqrt{196+144}$$

$$=\sqrt{340}$$

$$=2\sqrt{85}$$

Radius

$$=\frac{2\sqrt{85}}2=\sqrt{85}$$

VULMS Mastery

Step 3: Equation

$$
(x-1)^2+(y-4)^2=85
$$

Answer:

Center = **(1,4)**

Radius = **$\sqrt{85}$**

Equation:

$$
\boxed{(x-1)^2+(y-4)^2=85}
$$

Topic 7 – Trigonometry

Difficult Example 1 (Using Trigonometric Properties)

Question:

Given that

$$
\cos\theta=-\frac45
$$

and θ lies in the **second quadrant**, find:

a) $\sin\theta$

b) $\tan\theta$

Solution

Using the identity,

$$
\sin^2\theta+\cos^2\theta=1
$$

$$
\sin^2\theta=1-\left(\frac45\right)^2
$$

VULMS Mastery

$$
=1-\frac{16}{25}
$$

$$
=\frac9{25}
$$

Since θ is in Quadrant II,

$$
\sin\theta=\frac35
$$

Now,

$$
\tan\theta=\frac{\sin\theta}{\cos\theta}
$$

$$
=\frac{3/5}{-4/5}
$$

$$
=-\frac34
$$

Answer

$$
\boxed{\sin\theta=\frac35,\qquad \tan\theta=-\frac34}
$$

Difficult Example 2 (Amplitude and Period)

Question

Find the **amplitude** and **period** of

$$
y=-7\cos(4x)
$$

Solution

Amplitude

$$
=|A|=|-7|=7
$$

VULMS Mastery

Period

$$=\frac{360^\circ}{4}$$

$$=90^\circ$$

Answer

Amplitude = **7**

Period = **90°**

Difficult Example 3 (Graph Transformation)

Question

For the function

$$y=5\sin(3x)-2$$

determine

1.

Amplitude

2.
3.

Period

4.
5.

Vertical Shift

6.
7.

Direction of Shift

8.

VULMS Mastery

Solution

Amplitude

$$=5$$

Period

$$=\frac{360^\circ}{3}$$

$$=120^\circ$$

The "-2" shifts the graph downward by 2 units.

Answer

Amplitude = **5**

Period = **120°**

Vertical Shift = **2 units downward**

Difficult Example 4 (Phase Shift)

Question

Determine the phase shift of

$$y=\cos(x+75^\circ)$$

Solution

The rule is

$$\cos(x+k)$$

moves the graph **left** by (k).

Therefore,

VULMS Mastery

Shift

[
=75^\circ
]

Answer

[
\boxed{\text{75° to the left}}
]

Difficult Example 5 (Inverse Trigonometric Functions)

Question

Evaluate

a)

[
\arcsin\left(-\frac12\right)
]

b)

[
\arccos\left(-\frac{\sqrt3}{2}\right)
]

c)

[
\arctan(\sqrt3)
]

Solution

a)

[
\arcsin\left(-\frac12\right)

-\frac{\pi}{6}
]

b)

[
\arccos\left(-\frac{\sqrt3}{2}\right)

VULMS Mastery

\frac{5\pi}{6}
]

c)

[
\arctan(\sqrt3)

\frac{\pi}{3}
]

Answer

[
\boxed{
-\frac{\pi}{6},
\quad
\frac{5\pi}{6},
\quad
\frac{\pi}{3}
}
]

Difficult Example 6 (Solve Cos Equation)

Question

Solve

[
\cos\theta=-\frac12
]

for

[
0^\circ\le\theta\le360^\circ
]

Solution

Reference angle

[
60^\circ
]

Cosine is negative in Quadrants II and III.

Therefore,

VULMS Mastery

$$180^\circ-60^\circ=120^\circ$$

$$180^\circ+60^\circ=240^\circ$$

Answer

$$\boxed{\theta=120^\circ,\;240^\circ}$$

Difficult Example 7 (Solve Sine Equation)

Question

Solve

$$\sin\theta=\frac{\sqrt3}{2}$$

for

$$-180^\circ\le\theta\le180^\circ$$

Solution

Reference angle

$$60^\circ$$

Sine is positive in Quadrants I and II.

Solutions are

$$60^\circ$$

and

$$120^\circ$$

VULMS Mastery

Since the interval includes negative angles,

$$120^\circ-360^\circ=-240^\circ$$

which is outside the interval.

Hence only

$$60^\circ$$

Also,

$$180^\circ-60^\circ=120^\circ$$

Answer

$$\boxed{\theta=60^\circ,\;120^\circ}$$

VULMS Mastery

Difficult Example 8 (Solve Tangent Equation)

Question

Solve

$$\tan\theta=-1$$

for

$$0^\circ\le\theta\le360^\circ$$

Solution

Reference angle

$$45^\circ$$

Tangent is negative in Quadrants II and IV.

https://chat.whatsapp.com/Hrh5CrjNO6u31kFiNX352H

Therefore,

$$
180^\circ-45^\circ=135^\circ
$$

$$
360^\circ-45^\circ=315^\circ
$$

Answer

$$
\boxed{\theta=135^\circ,;315^\circ}
$$

Difficult Example 9 (Verify Identity)

Question

Verify the identity

$$
\frac{1-\cos^2\theta}{\sin\theta}
=\sin\theta
$$

Solution

LHS

$$
\frac{1-\cos^2\theta}{\sin\theta}
$$

Using

$$
1-\cos^2\theta=\sin^2\theta
$$

$$
\frac{\sin^2\theta}{\sin\theta}
$$

$$
=\sin\theta
$$

VULMS Mastery

LHS = RHS

Hence verified.

Answer

[
\boxed{\text{Identity Verified}}
]

Difficult Example 10 (Sum Formula)

Question

Find the exact value of

[
\sin75^\circ
]

Solution

Use

[
75^\circ=45^\circ+30^\circ
]

[
\sin(A+B)

\sin A\cos B+\cos A\sin B
]

[

\sin45^\circ\cos30^\circ
+
\cos45^\circ\sin30^\circ
]

Substitute values

[

\frac{\sqrt2}{2}\times\frac{\sqrt3}{2}
+
\frac{\sqrt2}{2}\times\frac12
]

[

VULMS Mastery

\frac{\sqrt6}{4}
+
\frac{\sqrt2}{4}
]

[

\frac{\sqrt6+\sqrt2}{4}
]

Answer

[
\boxed{\sin75^\circ=\frac{\sqrt6+\sqrt2}{4}}
]

Statistics (Topic 8)

Difficult Example 1 (Population, Sample, Parameter & Statistic)

Question

A university has **12,500 students**. A researcher randomly selects **400 students** and finds that **68% use online learning platforms daily**.

Identify:

a) Population

b) Sample

c) Parameter

d) Statistic

Solution

Population = All **12,500 students**

Sample = **400 selected students**

Parameter = Actual percentage of all 12,500 students using online learning platforms (unknown)

Statistic = **68%**, calculated from the sample

Answer

VULMS Mastery

Population = **12,500 students**

Sample = **400 students**

Parameter = **Actual population percentage**

Statistic = **68%**

Difficult Example 2 (Qualitative, Quantitative, Continuous & Discrete Variables)

Question

Classify each variable.

a) Number of books borrowed from a library

b) Student's blood pressure

c) Favorite social media platform

d) Daily rainfall (cm)

e) Number of cars in a parking lot

Solution

Number of books → Quantitative Discrete

Blood pressure → Quantitative Continuous

Favorite social media → Qualitative

Rainfall → Quantitative Continuous

Number of cars → Quantitative Discrete

Answer

Variable	Type
Books Borrowed	Quantitative (Discrete)
Blood Pressure	Quantitative (Continuous)
Favorite Social Media	Qualitative
Rainfall	Quantitative (Continuous)
Cars in Parking Lot	Quantitative (Discrete)

Difficult Example 3 (Identify Measurement Scale)

VULMS Mastery

Question

Identify the measurement scale for each variable.

a) Temperature in Celsius

b) Monthly Income

c) Employee Performance Rating (Excellent, Good, Fair, Poor)

d) Blood Group

e) Distance between two cities

Solution

Temperature (°C)

Difference matters but true zero does not.

→ Interval Scale

Monthly Income

Has a true zero.

→ Ratio Scale

Performance Rating

Ordered categories.

→ Ordinal Scale

Blood Group

Categories only.

→ Nominal Scale

Distance

Has a true zero.

→ Ratio Scale

Answer

Variable	Scale
Temperature	Interval
Monthly Income	Ratio

VULMS Mastery

Variable	Scale
Performance Rating	Ordinal
Blood Group	Nominal
Distance	Ratio

Difficult Example 4 (Parameter vs Statistic)

Question

A factory manufactures **18,000 bulbs**. Quality inspectors randomly test **600 bulbs** and find that **18 bulbs are defective**.

Find:

a) Sample defective percentage

b) Identify the statistic.

Solution

Sample Percentage

[
=\frac{18}{600}\times100
]

[
=3%
]

Since this value comes from the sample,

It is a **Statistic**.

VULMS Mastery

Answer

Sample Defective Percentage = **3%**

Statistic = **3%**

Difficult Example 5 (Inferential Statistics)

Question

A sample of **150 households** shows an average monthly electricity bill of **Rs. 7,800**.

Estimate the average bill for **20,000 households** using inferential statistics.

Solution

Inferential statistics use a sample to estimate the population.

Estimated population average

$$
=\text{Rs. }7,800
$$

Answer

Estimated average electricity bill

$$
\boxed{\text{Rs. }7,800}
$$

Difficult Example 6 (Descriptive Statistics)

Question

The marks of six students are

72, 65, 81, 94, 78, 70

Find

a) Total Marks

b) Average Marks

Solution

Total

$$
72+65+81+94+78+70
=460
$$

Average

$$
=\frac{460}{6}
$$

$$
=76.67
$$

VULMS Mastery

Answer

Total = **460**

Average = **76.67**

Difficult Example 7 (Sample Selection)

Question

A company has **2,400 employees**.

The HR department randomly selects **240 employees** for a health survey.

Find the sampling percentage.

Solution

[
\text{Sampling Percentage}

\frac{240}{2400}\times100
]

[
=10%
]

Answer

Sampling Percentage

[
\boxed{10%}
]

Difficult Example 8 (Measurement Scale Reasoning)

Question

Determine the measurement scale of each variable.

a) Jersey numbers of football players

b) IQ Scores

c) Body Weight

VULMS Mastery

d) Satisfaction Level (1–5)

Solution

Jersey Number

Numbers are labels only.

→ Nominal

IQ Score

Differences matter but true zero does not.

→ Interval

Body Weight

Has true zero.

→ Ratio

Satisfaction Level

Ordered categories.

→ Ordinal

Answer

Variable	Scale
Jersey Number	Nominal
IQ Score	Interval
Body Weight	Ratio
Satisfaction Level	Ordinal

Difficult Example 9 (Classifying Statistical Method)

Question

State whether each situation uses **Descriptive** or **Inferential Statistics**.

a) Average rainfall of the last 20 years is calculated.

b) Predicting next year's rainfall using historical data.

c) Reporting the average salary of company employees.

d) Predicting unemployment next year from survey data.

VULMS Mastery

Solution

Average rainfall

→ Descriptive

Prediction of rainfall

→ Inferential

Average salary

→ Descriptive

Prediction of unemployment

→ Inferential

Answer

Situation	Type
Average Rainfall	Descriptive
Rainfall Prediction	Inferential
Average Salary	Descriptive
Unemployment Prediction	Inferential

Difficult Example 10 (Comprehensive Statistics Problem)

Question

A school has **3,600 students**.

A researcher selects **300 students**.

Among them,

180 are boys

Average age = 15.6 years

Average height = 163 cm

Identify:

a) Population

b) Sample

VULMS Mastery

c) Statistic

d) Type of variables:

• Gender

• Age

• Height

Solution

Population

= All **3,600 students**

Sample

= **300 students**

Statistics

Average age = **15.6 years**

Average height = **163 cm**

Variable Types

Gender

→ Qualitative

Age

→ Quantitative Continuous

Height

→ Quantitative Continuous

Answer

Population = **3,600 students**

Sample = **300 students**

Statistics = **15.6 years, 163 cm**

Variable	**Type**
Gender	Qualitative
Age	Quantitative (Continuous)

VULMS Mastery

Variable	Type
Height	Quantitative (Continuous)

Representation of Data (Topic 9)

Difficult Example 1 (Constructing a Frequency Distribution)

Question

The marks of 20 students are:

42, 56, 73, 65, 81, 77, 69, 52, 48, 91,

84, 67, 59, 75, 63, 71, 54, 88, 60, 79

Construct a grouped frequency distribution using class intervals of width **10**.

VULMS Mastery

Solution

Marks	Frequency
40–49	2
50–59	4
60–69	5
70–79	5
80–89	3
90–99	1

Total Frequency

[
2+4+5+5+3+1=20
]

Answer

Class Interval	Frequency
40–49	2
50–59	4
60–69	5
70–79	5
80–89	3
90–99	1

Difficult Example 2 (Relative Frequency)

Question

Using the frequency distribution below, calculate the **relative frequency** of each class.

Class	**Frequency**
10–19	8
20–29	12
30–39	15
40–49	10
50–59	5

Solution

Total Frequency

[
8+12+15+10+5=50
]

Relative Frequency

10–19

[
\frac8{50}=0.16=16%
]

20–29

[
\frac{12}{50}=24%
]

30–39

[
\frac{15}{50}=30%
]

40–49

[
\frac{10}{50}=20%
]

50–59

[
\frac5{50}=10%
]

VULMS Mastery

Answer

Class	Relative Frequency
10–19	16%
20–29	24%
30–39	30%
40–49	20%
50–59	10%

Difficult Example 3 (Cumulative Frequency)

Question

Find the cumulative frequency.

Class	Frequency
5–14	4
15–24	9
25–34	12
35–44	10
45–54	5

Solution

Class	Frequency	Cumulative Frequency
5–14	4	4
15–24	9	13
25–34	12	25
35–44	10	35
45–54	5	40

VULMS Mastery

Answer

Cumulative Frequencies

[
4,;13,;25,;35,;40
]

Difficult Example 4 (Frequency Density)

Question

Calculate the frequency density.

Class	Frequency
20–30	12
30–45	18
45–60	24
60–80	20

Solution

Class Widths

20–30 → 10

30–45 → 15

45–60 → 15

60–80 → 20

Frequency Density

20–30

[
=\frac{12}{10}=1.2
]

30–45

[
=\frac{18}{15}=1.2
]

45–60

[
=\frac{24}{15}=1.6
]

60–80

[
=\frac{20}{20}=1
]

Answer

Class	Frequency Density
20–30	1.2
30–45	1.2
45–60	1.6

VULMS Mastery

Class	Frequency Density
60–80	1.0

Difficult Example 5 (Histogram)

Question

A histogram is to be drawn for the following data.

Class	Frequency
0–20	18
20–40	24
40–60	30
60–80	20

Find the height of each bar.

Solution

Since all class widths are equal,

Histogram Height

=

Frequency

Answer

Class	Bar Height
0–20	18
20–40	24
40–60	30
60–80	20

Difficult Example 6 (Unequal Class Width Histogram)

Question

Find the frequency density.

Class	Frequency
5–15	16
15–35	24
35–50	18
50–80	30

VULMS Mastery

Solution

Widths

10, 20, 15, 30

Frequency Density

[
\frac{16}{10}=1.6
]

[
\frac{24}{20}=1.2
]

[
\frac{18}{15}=1.2
]

[
\frac{30}{30}=1
]

Answer

Class	Frequency Density
5–15	1.6
15–35	1.2
35–50	1.2
50–80	1.0

Difficult Example 7 (Cumulative Frequency Application)

Question

The cumulative frequencies are

Upper Boundary	Cumulative Frequency
20	6
40	18
60	34
80	45
100	50

Find the percentage of observations **less than 60**.

Solution

VULMS Mastery

Frequency less than 60

[
=34
]

Total observations

[
=50
]

Percentage

[
=\frac{34}{50}\times100
]

[
=68%
]

Answer

[
\boxed{68%}
]

Difficult Example 8 (Pie Chart Angles)

Question

A survey of **240 students** gives the following results.

Activity	Frequency
Reading	72
Sports	60
Music	48
Gaming	36
Others	24

Calculate the angle for each sector of the pie chart.

Solution

Formula

[
\text{Angle}

VULMS Mastery

\frac{\text{Frequency}}{\text{Total}}
\times360^\circ
]

Reading

[
=\frac{72}{240}\times360
=108^\circ
]

Sports

[
=90^\circ
]

Music

[
=72^\circ
]

Gaming

[
=54^\circ
]

Others

[
=36^\circ
]

Answer

Activity	**Angle**
Reading	108°
Sports	90°
Music	72°
Gaming	54°
Others	36°

Difficult Example 9 (Reverse Pie Chart)

Question

A pie chart has a sector measuring **81°**.

If the total number of observations is **160**, find the frequency represented by the sector.

https://chat.whatsapp.com/Hrh5CrjNO6u31kFiNX352H

Solution

Formula

[
\text{Frequency}

\frac{\theta}{360}\times n
]

[

\frac{81}{360}\times160
]

[
=36
]

Answer

[
\boxed{36}
]

Difficult Example 10 (Comprehensive Data Representation)

Question

The grouped data below represent examination marks.

Marks	Frequency
0–20	6
20–40	14
40–60	18
60–80	10
80–100	2

Find:

a) Total observations

b) Relative frequency of 40–60

c) Cumulative frequency up to 60

d) Pie chart angle for 60–80

Solution

VULMS Mastery

https://chat.whatsapp.com/Hrh5CrjNO6u31kFiNX352H

Step 1

Total

[
6+14+18+10+2
=50
]

Step 2

Relative Frequency

[
=\frac{18}{50}
=0.36
=36%
]

Step 3

Cumulative Frequency

[
6+14+18
=38
]

Step 4

Pie Chart Angle

[
=\frac{10}{50}\times360
]

[
=72^\circ
]

Answer

Total Observations = **50**

Relative Frequency (40–60) = **36%**

Cumulative Frequency (up to 60) = **38**

Pie Chart Angle (60–80) = **72°**

VULMS Mastery

MTH100 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Measures of Central Tendency (Topic 10)

Difficult Example 1 (Arithmetic Mean – Ungrouped Data)

Question

The monthly sales (in thousands of rupees) of a shop for eight months are:

85, 92, 76, 88, 95, 81, 79, 104

Find the arithmetic mean.

Solution

Formula

[
\bar{x}=\frac{\sum x}{n}
]

Sum

[
85+92+76+88+95+81+79+104=700
]

Number of observations

[
n=8
]

Mean

[
\bar{x}=\frac{700}{8}=87.5
]

Answer

[
\boxed{\bar{x}=87.5}
]

Difficult Example 2 (Arithmetic Mean – Grouped Data)

Question

VULMS Mastery

Find the mean of the following grouped data.

Class	Frequency
10–20	5
20–30	8
30–40	10
40–50	7

Solution

First find the class midpoints.

Class	Midpoint (z)	Frequency (f)	fz
10–20	15	5	75
20–30	25	8	200
30–40	35	10	350
40–50	45	7	315

[
\sum f=30
]

[
\sum fz=940
]

Mean

[
\bar{x}=\frac{\sum fz}{\sum f}
]

[
=\frac{940}{30}
]

[
=31.33
]

Answer

[
\boxed{31.33}
]

Difficult Example 3 (Geometric Mean)

Question

VULMS Mastery

An investment grows by **8%, 12%, and 15%** over three consecutive years.

Find the average annual growth rate.

Solution

Formula

[
GM=\sqrt[3]{(1.08)(1.12)(1.15)}
]

[
=\sqrt[3]{1.39104}
]

[
=1.1163
]

Growth rate

[
=(1.1163-1)\times100
]

[
=11.63%
]

Answer

[
\boxed{11.63%}
]

VULMS Mastery

Difficult Example 4 (Weighted Mean)

Question

Three sections have the following average marks.

Section	Students	Average Marks
A	25	72
B	40	81
C	35	78

Find the overall average marks.

Solution

Formula

[
\bar{x}

\frac{\sum wx}{\sum w}
]

Weighted total

[
25(72)+40(81)+35(78)
]

[
=1800+3240+2730
]

[
=7770
]

Total students

[
25+40+35=100
]

Overall Mean

[
=\frac{7770}{100}
]

[
=77.7
]

Answer

[
\boxed{77.7}
]

Difficult Example 5 (Harmonic Mean)

Question

A cyclist travels **60 km** at **30 km/h** and another **60 km** at **45 km/h**.

Find the average speed.

VULMS Mastery

Solution

Since equal distances are covered,

Use Harmonic Mean.

[
HM

\frac{2}
{\frac1{30}+\frac1{45}}
]

LCM = 90

[
=\frac2{\frac3{90}+\frac2{90}}
]

[
=\frac2{\frac5{90}}
]

[
=\frac{180}{5}
]

[
=36
]

Answer

[
\boxed{36\text{ km/h}}
]

VULMS Mastery

Difficult Example 6 (Median – Odd Number of Observations)

Question

Find the median.

45, 60, 52, 71, 68, 49, 80, 56, 63

Solution

Arrange in ascending order.

45, 49, 52, 56, 60, 63, 68, 71, 80

https://chat.whatsapp.com/Hrh5CrjNO6u31kFiNX352H

Number of observations

[
n=9
]

Median Position

[
=\frac{9+1}{2}=5
]

5th observation

[
=60
]

Answer

[
\boxed{60}
]

Difficult Example 7 (Median – Even Number of Observations)

Question

Find the median.

38, 54, 42, 61, 58, 47, 65, 50

Solution

Arrange

38, 42, 47, 50, 54, 58, 61, 65

There are

[
8
]

observations.

Median

VULMS Mastery

[
=\frac{50+54}{2}
]

[
=52
]

Answer

[
\boxed{52}
]

Difficult Example 8 (Median – Grouped Data)

Question

Find the median.

Class	Frequency
0–10	4
10–20	8
20–30	14
30–40	10
40–50	4

Solution

Total Frequency

[
N=40
]

[
N/2=20
]

Cumulative Frequency

Class	CF
0–10	4
10–20	12
20–30	26
30–40	36
40–50	40

VULMS Mastery

Median Class

20–30

Here

[
L=20
]

[
F=12
]

[
f=14
]

[
h=10
]

Formula

[
Median

20+
\left(
\frac{20-12}{14}
\right)
\times10
]

[
=20+5.71
]

[
=25.71
]

Answer

[
\boxed{25.71}
]

Difficult Example 9 (Mode – Grouped Data)

Question

VULMS Mastery

MTH100 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Find the mode.

Class	Frequency
10–20	6
20–30	12
30–40	18
40–50	15
50–60	9

Solution

Modal Class

30–40

[
L=30
]

[
h=10
]

[
\Delta_1=18-12=6
]

[
\Delta_2=18-15=3
]

Formula

[
Mode

30+
\frac6{6+3}
\times10
]

[
=30+6.67
]

[
=36.67
]

Answer

VULMS Mastery

https://chat.whatsapp.com/Hrh5CrjNO6u31kFiNX352H

[
\boxed{36.67}
]

Difficult Example 10 (Comprehensive Central Tendency Problem)

Question

The marks obtained by ten students are

48, 55, 62, 62, 70, 75, 75, 75, 82, 96

Find:

a) Arithmetic Mean

b) Median

c) Mode

Solution

Step 1

Sum

[
48+55+62+62+70+75+75+75+82+96
=700
]

Mean

[
=\frac{700}{10}
=70
]

Step 2

Median

Middle values are

70 and 75

[
Median

VULMS Mastery

\frac{70+75}{2}
=72.5
]

Step 3

Mode

75 appears **3 times**, more than any other value.

Mode

[
=75
]

Answer

Arithmetic Mean = **70**

Median = **72.5**

Mode = **75**

Measures of Dispersion (Topic 11)

Difficult Example 1 (Range)

Question

The daily temperatures (°C) recorded over eight days are:

18, 25, 31, 29, 22, 34, 27, 20

Find the **Range**.

Solution

Largest Value

[
=34
]

Smallest Value

VULMS Mastery

[
=18
]

Range

[
=34-18
]

[
=16
]

Answer

[
\boxed{\text{Range}=16}
]

Difficult Example 2 (Percentiles)

Question

The following data are arranged in ascending order:

12, 15, 18, 22, 24, 27, 30, 35, 38, 41, 45, 49

Find the:

a) 25th Percentile

b) 75th Percentile

Solution

Number of observations

[
n=12
]

Location Formula

[
L_y=(n+1)\times\frac{y}{100}
]

Part (a)

VULMS Mastery

[
L_{25}

13\times\frac{25}{100}
]

[
=3.25
]

Approximate 25th percentile

[
=P_{25}=18
]

Part (b)

[
L_{75}

13\times\frac{75}{100}
]

[
=9.75
]

Approximate value

[
=P_{75}=38
]

Answer

[
\boxed{P_{25}=18,\qquad P_{75}=38}
]

Difficult Example 3 (Quartiles and IQR)

Question

Find

a) (Q_1)

b) Median

c) (Q_3)

VULMS Mastery

d) Interquartile Range

for

6, 8, 11, 13, 16, 18, 20, 22, 25, 27, 30

Solution

Number of observations

[
n=11
]

Median

6th value

[
=18
]

Remove the median.

Lower Half

6, 8, 11, 13, 16

Upper Half

20, 22, 25, 27, 30

Lower Median

[
Q_1=11
]

Upper Median

[
Q_3=25
]

IQR

[
=25-11
]

VULMS Mastery

[
=14
]

Answer

[
\boxed{Q_1=11,;Q_2=18,;Q_3=25,;IQR=14}
]

Difficult Example 4 (Five Number Summary)

Question

Find the five-number summary.

3, 5, 8, 10, 12, 15, 17, 20, 24

Solution

Minimum

[
=3
]

Median

5th value

[
=12
]

Lower Half

3,5,8,10

[
Q_1

\frac{5+8}{2}
=6.5
]

Upper Half

15,17,20,24

VULMS Mastery

[
Q_3

\frac{17+20}{2}
=18.5
]

Maximum

[
=24
]

Answer

Minimum = **3**

[
Q_1=6.5
]

Median = **12**

[
Q_3=18.5
]

Maximum = **24**

Difficult Example 5 (Mean Absolute Deviation)

Question

Find the Mean Absolute Deviation (MAD).

12, 18, 20, 24, 26

Solution

Mean

[
=\frac{12+18+20+24+26}{5}
]

[
=\frac{100}{5}
=20
]

Absolute Deviations

VULMS Mastery

Observation	(|x-\bar x|)
12	8
18	2
20	0
24	4
26	6

Sum

[
8+2+0+4+6=20
]

MAD

[
=\frac{20}{5}
=4
]

Answer

[
\boxed{MAD=4}
]

Difficult Example 6 (Variance and Standard Deviation)

Question

Find the population variance and standard deviation.

5, 7, 9, 11, 13

Solution

Mean

[
=\frac{45}{5}
=9
]

Squared Deviations

Observation	((x-\bar x)^2)
5	16
7	4
9	0

VULMS Mastery

Observation	**((x-\bar x)^2)**
11	4
13	16

Sum

[
16+4+0+4+16
=40
]

Variance

[
=\frac{40}{5}
=8
]

Standard Deviation

[
=\sqrt8
]

[
=2.83
]

Answer

Variance

[
\boxed{8}
]

Standard Deviation

[
\boxed{2.83}
]

Difficult Example 7 (Sample Variance)

Question

A sample consists of

14, 16, 18, 20, 22

VULMS Mastery

Find the unbiased sample variance.

Solution

Mean

[
=18
]

Squared Deviations

16,4,0,4,16

Sum

[
40
]

Sample Variance

[
s^2

\frac{40}{5-1}
]

[
=10
]

Sample Standard Deviation

[
=\sqrt{10}
=3.16
]

Answer

[
\boxed{s^2=10,\qquad s=3.16}
]

Difficult Example 8 (Variance of Grouped Data)

Question

Find the variance.

VULMS Mastery

Class	Frequency
10–20	4
20–30	6
30–40	8
40–50	2

Solution

Midpoints

15,25,35,45

Midpoint	Frequency	(fx)	(fx^2)
15	4	60	900
25	6	150	3750
35	8	280	9800
45	2	90	4050

[
\sum f=20
]

[
\sum fx=580
]

Mean

[
=29
]

[
\sum fx^2=18500
]

Variance

[

\frac{18500}{20}

29^2
]

[
=925-841
]

VULMS Mastery

[
=84
]

Answer

[
\boxed{\text{Variance}=84}
]

Difficult Example 9 (Chebyshev's Theorem)

Question

A distribution has mean

[
50
]

and standard deviation

[
8
]

Find the minimum percentage of observations lying within **3 standard deviations**.

Solution

Chebyshev's Formula

[
1-\frac1{k^2}
]

Here

[
k=3
]

[
1-\frac19
]

[
=\frac89
]

VULMS Mastery

[
=88.89%
]

Answer

[
\boxed{88.89%}
]

Difficult Example 10 (Coefficient of Variation)

Question

Investment A

Mean Return = 15%

Standard Deviation = 4%

Investment B

Mean Return = 10%

Standard Deviation = 2.8%

Find the coefficient of variation and determine which investment is riskier.

Solution

Formula

[
CV=\frac{s}{\bar x}
]

Investment A

[
CV=\frac4{15}
=0.267
]

Investment B

[
CV=\frac{2.8}{10}
=0.280
]

VULMS Mastery

Since

[
0.280>0.267
]

Investment B has greater relative risk.

Answer

Investment A

[
\boxed{CV=0.267}
]

Investment B

[
\boxed{CV=0.280}
]

Riskier Investment: Investment B

Topic 12 – Skewness and Kurtosis

Example 1: Find the Median using Mean and Mode

Question:

The mean of a distribution is **42** and the mode is **36**. Find the median using the empirical relation.

Solution:

Formula:

Mode = 3 Median − 2 Mean

36 = 3Median − 2(42)

36 = 3Median − 84

3Median = 120

Median = 40

Answer: Median = **40**

VULMS Mastery

Example 2: Find the Mean

Question:

The mode of a distribution is **50** and the median is **46**. Find the mean.

Solution:

Mode = 3Median − 2Mean

50 = 3(46) − 2Mean

50 = 138 − 2Mean

2Mean = 88

Mean = 44

Answer: Mean = **44**

Example 3: Pearson's Coefficient of Skewness (Using Mode)

Question:

Mean = 72

Mode = 66

Standard deviation = 12

Find Pearson’s coefficient of skewness.

Solution:

[
\text{Skewness}=\frac{\text{Mean}-\text{Mode}}{\text{SD}}
]

[
=\frac{72-66}{12}
=\frac{6}{12}
=0.50
]

Answer: Pearson's coefficient = **0.50 (Positively Skewed)**

VULMS Mastery

https://chat.whatsapp.com/Hrh5CrjNO6u31kFiNX352H

Example 4: Pearson's Coefficient (Using Median)

Question:

Mean = 55

Median = 52

Standard deviation = 9

Find Pearson’s coefficient of skewness.

Solution:

[
\text{Skewness}=\frac{3(\text{Mean}-\text{Median})}{SD}
]

[
=\frac{3(55-52)}{9}
=\frac{9}{9}
=1
]

Answer: Pearson's coefficient = **1 (Highly Positively Skewed)**

Example 5: Determine the Type of Skewness

Question:

Mean = 38

Median = 40

Mode = 43

Determine whether the distribution is positively or negatively skewed.

Solution:

Since

Mean < Median < Mode

38 < 40 < 43

This represents a **negative (left) skewed distribution.**

Answer: Negatively Skewed

VULMS Mastery

Example 6: Calculate the Second Central Moment

Question:

A data set has values:

6, 8, 10, 12, 14

Find the second central moment (μ_2).

Solution:

Mean

[
=\frac{6+8+10+12+14}{5}=10
]

x	x−Mean	(x−Mean)²
6	-4	16
8	-2	4
10	0	0
12	2	4
14	4	16

Sum = 40

[
\mu_2=\frac{40}{5}=8
]

Answer: $\mu_2 = 8$

Example 7: Calculate the Third Central Moment

Question:

Using the same data set

6, 8, 10, 12, 14

Find μ_3.

Solution:

x	x−Mean	(x−Mean)³

VULMS Mastery

x	x−Mean	(x−Mean)³
6	-4	-64
8	-2	-8
10	0	0
12	2	8
14	4	64

Sum

= -64 -8 +0 +8 +64

=0

$$\mu_3=\frac{0}{5}=0$$

Answer: $\mu_3 = 0$ (Perfectly Symmetric)

Example 8: Find Moment Coefficient of Skewness

Question:

Given

$\mu_2 = 16$

$\mu_3 = 32$

Find β_1.

Solution:

Formula

$$\beta_1=\frac{\mu_3^2}{\mu_2^3}$$

$$=\frac{32^2}{16^3}
=\frac{1024}{4096}
=0.25$$

Since β_1 is positive,

the distribution is **positively skewed**.

VULMS Mastery

https://chat.whatsapp.com/Hrh5CrjNO6u31kFiNX352H

Answer:

β_1 = **0.25**

Distribution = **Positively Skewed**

Example 9: Calculate Kurtosis

Question:

Given

$\mu_2 = 9$

$\mu_4 = 405$

Find β_2 and identify the distribution.

Solution:

Formula

[
\beta_2=\frac{\mu_4}{\mu_2^2}
]

[
=\frac{405}{9^2}
=\frac{405}{81}
=5
]

Since

$\beta_2 > 3$

The distribution is **Leptokurtic**.

Answer:

β_2 = **5**

Distribution = **Leptokurtic**

Example 10: Identify the Type of Kurtosis

Question:

VULMS Mastery

For each value of β_2 determine the type of distribution.

(a) $\beta_2 = 2.6$

(b) $\beta_2 = 3$

(c) $\beta_2 = 4.8$

Solution:

(a)

$\beta_2 = 2.6$

Since $\beta_2 < 3$

Platykurtic

(b)

$\beta_2 = 3$

Since $\beta_2 = 3$

Mesokurtic

(c)

$\beta_2 = 4.8$

Since $\beta_2 > 3$

Leptokurtic

Answers

(a) **Platykurtic**

(b) **Mesokurtic**

(c) **Leptokurtic**

Topic 13 – Probability Concepts

Example 1: Probability of Drawing a Card

VULMS Mastery

Question:

A standard deck contains **52 cards**. Find the probability of drawing a **King**.

Solution:

Number of Kings = **4**

Total cards = **52**

$$
P(\text{King})=\frac{4}{52}=\frac{1}{13}
$$

Answer: 1/13

Example 2: Probability of Getting Exactly Two Heads

Question:

A fair coin is tossed **3 times**. Find the probability of getting **exactly two heads**.

Solution:

Sample Space

S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}

Total outcomes = **8**

Favorable outcomes

HHT, HTH, THH

Number of favorable outcomes = **3**

$$
P(A)=\frac{3}{8}
$$

Answer: 3/8

Example 3: Complement Rule

Question:

A bag contains **12 red balls** and **8 blue balls**.

VULMS Mastery

Find the probability of **not selecting a red ball**.

Solution:

Total balls

=12+8=20

Probability of Red

[
P(R)=\frac{12}{20}=\frac35
]

Complement

[
P(R')=1-\frac35=\frac25
]

Answer: 2/5

Example 4: Addition Rule

Question:

A die is rolled once.

Find the probability of getting **an even number or a number greater than 4**.

Solution:

Even numbers

{2,4,6}

[
P(A)=\frac36
]

Numbers greater than 4

{5,6}

[
P(B)=\frac26
]

Common number

VULMS Mastery

{6}

[
P(A\cap B)=\frac16
]

Addition Rule

[
P(A\cup B)=\frac36+\frac26-\frac16
=\frac46
=\frac23
]

Answer: 2/3

Example 5: Multiplication Rule

Question:

Two fair coins are tossed.

Find the probability of getting **Head on both coins**.

Solution:

Probability of Head on first coin

[
=\frac12
]

Probability of Head on second coin

[
=\frac12
]

Multiplication Rule

[
P(HH)=\frac12\times\frac12=\frac14
]

Answer: 1/4

Example 6: Independent Events

VULMS Mastery

Question:

A die is rolled and a coin is tossed.

Find the probability of getting **an odd number and a head**.

Solution:

Odd numbers

{1,3,5}

[
P(\text{Odd})=\frac36=\frac12
]

Head

[
P(H)=\frac12
]

Since both events are independent,

[
P=\frac12\times\frac12=\frac14
]

Answer: 1/4

Example 7: Sample Space Problem

VULMS Mastery

Question:

Two dice are rolled.

Find the probability that the **sum equals 9**.

Solution:

Total outcomes

[
6\times6=36
]

Sum = 9

(3,6)

(4,5)

(5,4)

(6,3)

Number of favorable outcomes

=4

$$
P=\frac4{36}=\frac19
$$

Answer: 1/9

Example 8: Mutually Exclusive Events

Question:

A card is drawn from a deck of 52 cards.

Find the probability of drawing **a Queen or a King**.

Solution:

Queens = 4

Kings = 4

A card cannot be both Queen and King.

Therefore,

$$
P(Q\cap K)=0
$$

Addition Rule

$$
P(Q\cup K)=\frac4{52}+\frac4{52}
=\frac8{52}
=\frac2{13}
$$

Answer: 2/13

VULMS Mastery

Example 9: Three Independent Events

Question:

A coin is tossed three times.

Find the probability of getting **Head, Tail, Head (HTH)**.

Solution:

Each toss

$$
P=\frac12
$$

Multiplication Rule

$$
P(HTH)=\frac12\times\frac12\times\frac12
=\frac18
$$

Answer: 1/8

Example 10: Combined Addition and Complement Rule

Question:

In a class of **50 students**,

30 study Mathematics,

25 study Physics,

15 study both.

Find the probability that a randomly selected student studies **neither Mathematics nor Physics**.

Solution:

Using Addition Rule

$$
P(M\cup P)=\frac{30+25-15}{50}
=\frac{40}{50}
=\frac45
$$

Using Complement Rule

VULMS Mastery

[
P(\text{Neither})
=1-\frac45
=\frac15
]

Answer: 1/5

⭑ **Bonus (Exam-Level Difficult)**

Example 11: Probability Using Sample Space

Question:

Three fair coins are tossed.

Find the probability of getting **at least two heads**.

Solution:

Sample Space

{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}

At least two heads means

HHH

HHT

HTH

THH

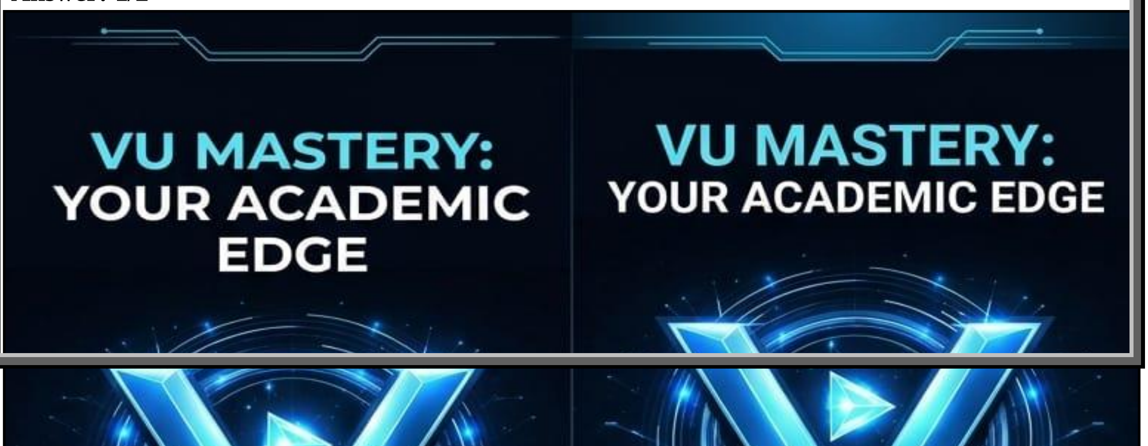
Total favorable outcomes = **4**

Total outcomes = **8**

[
P=\frac48=\frac12
]

Answer: 1/2

VULMS Mastery



MTH100 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

VULMS Mastery